

AUTO DATA STORYTELLER REPORT

Executive Summary

Executive Summary Our recent analytical initiative has produced a highly effective predictive model that accurately forecasts outcomes for over 83% of cases. This strong performance indicates a robust understanding of the underlying factors influencing the predicted event, providing a reliable foundation for informed strategic planning and operational adjustments. The analysis highlights three major business drivers significantly impacting the outcome. Specifically, an individual's gender (male), the fare paid, and age are identified as the most influential factors. These variables act as critical indicators, shaping the likelihood of a particular outcome and representing key segments or conditions the business should monitor closely. Given the substantial influence of these demographic and financial attributes, we strongly recommend continuous tracking of their trends and fluctuations. Changes in these primary drivers are likely to have the largest effect on future outcomes. Leveraging these reliable insights will be crucial for anticipating market shifts, optimizing resource allocation, and developing targeted strategies to effectively manage or influence the predicted outcomes.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

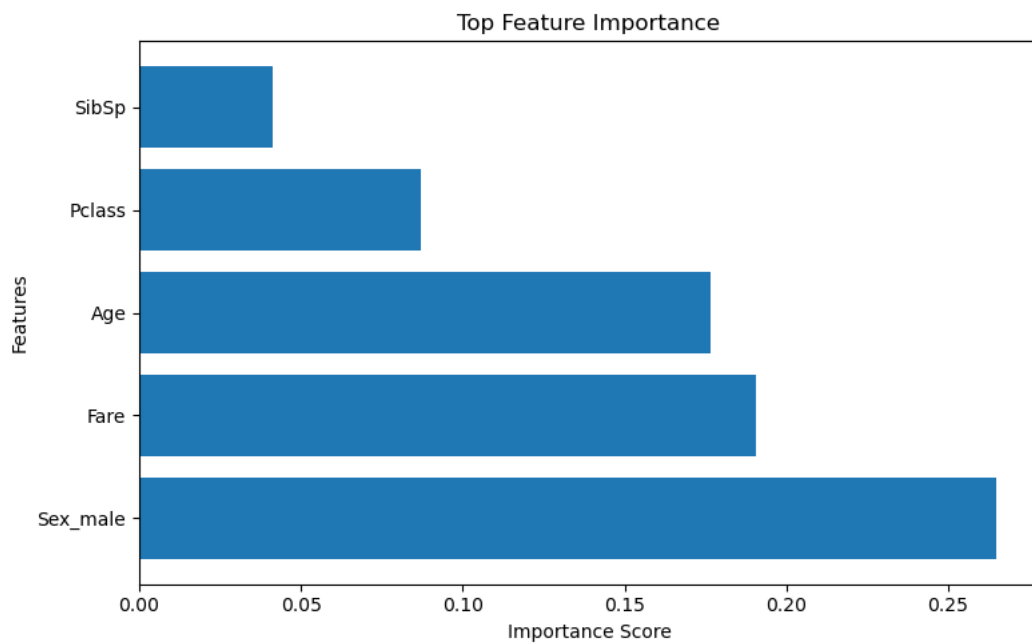
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

